

Automated Employee Onboarding & Off boarding System

Project Description:

The Automated Employee Onboarding & Offboarding System is a ServiceNow-based employee lifecycle management solution designed to digitize and automate the complete process of employee joining and exit activities. The system enables organizations to manage onboarding and offboarding requests through centralized Service Catalog items, automated approval workflows, departmental task assignments, SLA tracking, and secure role-based access.

The solution eliminates manual coordination between HR, Managers, IT, Facilities, and Security teams by using ServiceNow Flow Designer to automate request processing and task orchestration. Employees or managers can submit onboarding/offboarding requests through the Service Portal by providing required details such as Employee ID, Department, Manager, Joining Date, Exit Date, Asset Requirements, and Access Details.

Submitted requests are automatically stored in the Employee Lifecycle custom table (u_employee_lifecycle) for structured tracking. Flow Designer processes the request by routing approvals to managers and generating fulfillment tasks for responsible teams.

Scenarios

Scenario 1: New Employee Onboarding

A manager submits an onboarding request for a new employee through the ServiceNow Service Portal.

The user provides:

- Employee ID
- Employee Name
- Department
- Manager Details
- Joining Date
- Required Assets
- System Access Requirements

Scenario 2: Employee Offboarding

The workflow automatically creates tasks:

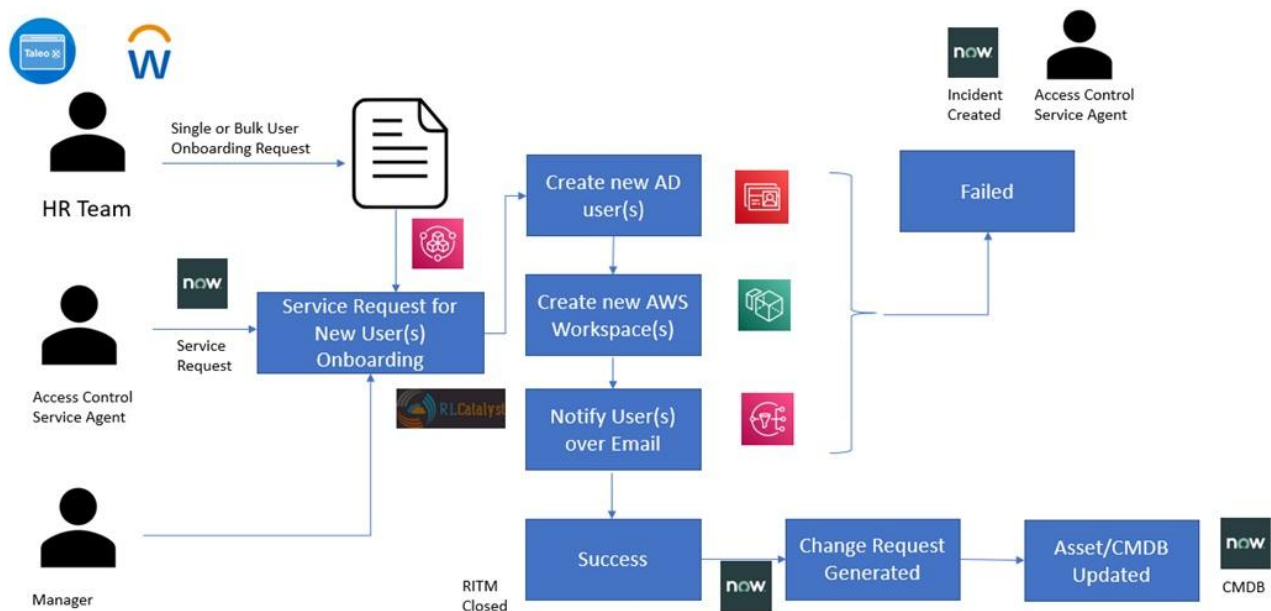
- IT disables system access and retrieves assets.
- Facilities collects workplace assets and access cards.
- Security revokes physical access.
- HR completes exit formalities.

Scenario 3 : Compliance and Reporting

* Number of onboarding requests

* Number of offboarding requests

Technical Architecture:



Description :

The Automated Employee Onboarding & Offboarding System is built on the ServiceNow platform using a modular architecture that integrates user interaction, data management, workflow automation, security controls, and reporting capabilities. The architecture is designed to streamline employee lifecycle processes by connecting HR, Managers, IT, Facilities, and Security teams through automated workflows.

Pre-requisites:

The following knowledge and tools are required:

ServiceNow Knowledge

- Service Catalog
- Flow Designer
- Custom Tables
- ACL Configuration
- Notifications
- SLA Management
- Reports and Dashboards

Project Workflow:

Milestone 1: Requirement Analysis and Solution Architecture

This milestone focuses on understanding business requirements, identifying stakeholders, and designing the overall ServiceNow solution architecture for automating employee lifecycle processes.

Activity 1.1: Analyze Employee Lifecycle Requirements

- Identify current challenges in manual onboarding and offboarding processes.

- Define business objectives such as:
 - Reducing manual coordination
 - Improving employee experience
 - Automating approvals and task assignments
 - Ensuring secure access management
 - Supporting compliance requirements
- Identify key stakeholders:
 - HR Team
 - Managers
 - IT Team
 - Facilities Team
 - Security Team

Variables (8)

Variable Sets

Catalog UI Policies

Catalog Client Scripts

Available For

Not Available For

Categories (1)

Catalogs (1)

Catalog Data Lookup Definitions

Related Articles

Related Catalog Items

Assigned Topics

≡

▽

⌵

Order

Search

⌵

—

Actions on selected rows...

New

Catalog item = Employee Onboarding Request

🔍 Type	Question	Order	⬆
Single Line Text	Employee Name	100	
Email	Email	200	
Single Line Text	Department	300	
Single Line Text	Role	400	
⌵ Reference	Manager	500	
Date	Joining Date	600	
Yes / No	Laptop Required	700	
Yes / No	System Access Required	800	

⏪

⏩

1 to 8 of 8

⏪

⏩

Steps to Generate API Key:

1. Open the Groq Console:
 - Navigate to the Groq developer platform.
 - Login using a registered account.
2. Access API Keys:
 - From the Groq dashboard, navigate to the **API Keys** section.
 - The API Keys page provides options to create and manage API credentials.
3. Create a New API Key:
 - Click on the **Create API Key** option.
 - Provide a meaningful name for the key, such as:

CaptionAI-Key

- Generate the API key.
4. Secure API Key Storage:
 - Copy the generated API key immediately.
 - Store it securely in the project environment file.

Example:

GROQ_API_KEY=your_generated_api_key_here

Activity 1.2: Design ServiceNow Solution Architecture

- Define the architecture components required for the solution:
 - Service Portal

- Service Catalog
- Employee Lifecycle Custom Table
- Flow Designer Automation
- ACL Security
- SLA Configuration
- Reports and Dashboards
- Design the process flow:

Employee Request → Catalog Submission → Manager Approval → Department Tasks → SLA Tracking → Completion → Lifecycle Closure

Activity 1.3: Define Data Model and Security Approach

- Create the Employee Lifecycle data model.

Custom Table:

u_employee_lifecycle

Store employee information:

- Employee ID
- Employee Name
- Request Type
- Department
- Manager
- Joining Date
- Exit Date
- Assets
- Access Details
- Status
- Define role-based security requirements using ACLs.

Activity 1.4: Configure ServiceNow Development Environment

- Set up ServiceNow Personal Developer Instance (PDI).
- Configure required application modules.
- Prepare development environment for:
 - Tables
 - Catalog Items
 - Variables
 - Flows
 - ACLs
 - Notifications
 - Reports

Milestone 2: Core Functionalities Development

This milestone focuses on developing the primary employee lifecycle capabilities using ServiceNow configuration components.

Activity 2.1: Develop Employee Onboarding and Offboarding Catalog Items

Create Service Catalog items:

Employee Onboarding

Capture details:

- **Employee ID**
- **Employee Name**
- **Department**
- **Manager**
- **Joining Date**
- **Required Assets**
- **System Access Requirements**

Employee Offboarding

Capture details:

- **Employee ID**
- **Department**
- **Manager**
- **Last Working Date**
- **Asset Return Details**
- **Access Revocation Details**

Activity 2.2: Configure Dynamic Variables and Form Behavior

- **Create catalog variables for capturing employee information.**
- **Implement UI Policies for dynamic field visibility.**

Examples:

When Onboarding is selected:

- **Display Joining Date**
- **Asset Requirements**
- **Access Requirements**

When Offboarding is selected:

- **Display Exit Date**
- **Asset Return Details**
- **Access Revocation Details**
- **Configure mandatory fields:**
 - **Employee ID**
 - **Request Type**
 - **Department**
 - **Manager**

Milestone 3: Flow Designer Automation Development

This milestone focuses on creating automated workflows for employee lifecycle management.

Activity 3.1: Create Employee Lifecycle Workflow

Develop Flow Designer workflows to automate request processing.

Flow Trigger:

When Employee Onboarding/Offboarding Catalog Item is submitted

Workflow Steps:

1. Retrieve catalog variables.
2. Create/update Employee Lifecycle record.
3. Send request for manager approval.
4. Validate approval outcome.
5. Generate departmental tasks.
6. Send email notifications.
7. Update lifecycle status.

Activity 3.2: Implement Approval and Task Automation

Configure dynamic approval routing:

Manager Approval → Department Fulfillment

Automatically create tasks for:

HR Team

- Employee documentation
- Lifecycle processing

IT Team

- User account creation/removal
- Application access
- Asset allocation/recovery

Facilities Team

- Workspace preparation
- Access card management

Security Team

- Physical access provisioning/revocation

Milestone 4: Security, Interface and Reporting Configuration

This milestone focuses on securing the application, improving usability, and enabling process monitoring.

Activity 4.1: Implement Role-Based Access Control

Configure ACLs and roles.

Roles:

- HR User
- Manager
- IT Fulfillment User
- Facilities User
- Security User
- Administrator

Access control ensures:

- HR manages employee requests.
- Managers approve employee requests.
- IT, Facilities, and Security access only assigned tasks.
- Sensitive employee data remains protected.

Activity 4.2: Configure SLA and Notifications

Configure SLA tracking for departmental tasks.

Monitor:

- Task completion time
- Approval delays
- Fulfillment timelines

Configure automated email notifications for:

- Request submission
- Approval pending
- Task assignment
- Task completion
- Request closure

Activity 4.3: Create Reports and Dashboards

Develop dashboards displaying:

- Total onboarding requests
- Total offboarding requests
- Pending approvals
- SLA compliance
- Task completion rates
- Lifecycle status tracking

Milestone 5: Testing, Deployment and Documentation

This milestone validates the complete solution and prepares it for deployment.

Activity 5.1: Perform QA Testing

Execute testing scenarios:

Request Lifecycle Testing

Validate:

- Request creation
- Variable mapping
- Employee Lifecycle record creation

Workflow Testing

Verify:

- Manager approval routing
- Task generation

- Status updates

Notification Testing

Verify:

- Email notifications
- Approval messages
- Completion alerts

Security Testing

Validate:

- ACL restrictions
- Role-based access

Activity 5.2: Deployment and Documentation

Prepare deployment deliverables:

- Update Sets
- Configuration documentation
- Workflow screenshots
- Table schema documentation
- Variable mapping details
- Test results
- User guide

Deploy configurations between ServiceNow instances using update sets.

Milestone 6: Conclusion

The **Automated Employee Onboarding & Offboarding System in ServiceNow** successfully digitizes employee lifecycle management by replacing manual processes with automated workflows. The solution improves collaboration between HR, Managers, IT, Facilities, and Security teams through centralized request management, automated approvals, task assignments, SLA monitoring, and secure access controls.

Future enhancements can include integrations with HR Management Systems, Identity Management platforms, and Asset Management systems to achieve complete end-to-end employee lifecycle automation.

Exploring web pages:

Variable Manager

Application: Global

Map to field: ☐

Type: Reference

Catalog Item: Employee Onboarding Request

Order: 500

Active: ☒

Mandatory: ☒

Read only: ☐

Hidden: ☐

Unique: ☐

Disable automatic slot fill based on user context: ☐

Question

Annotation

Type Specifications

Default Value

Auto-populate

Permission

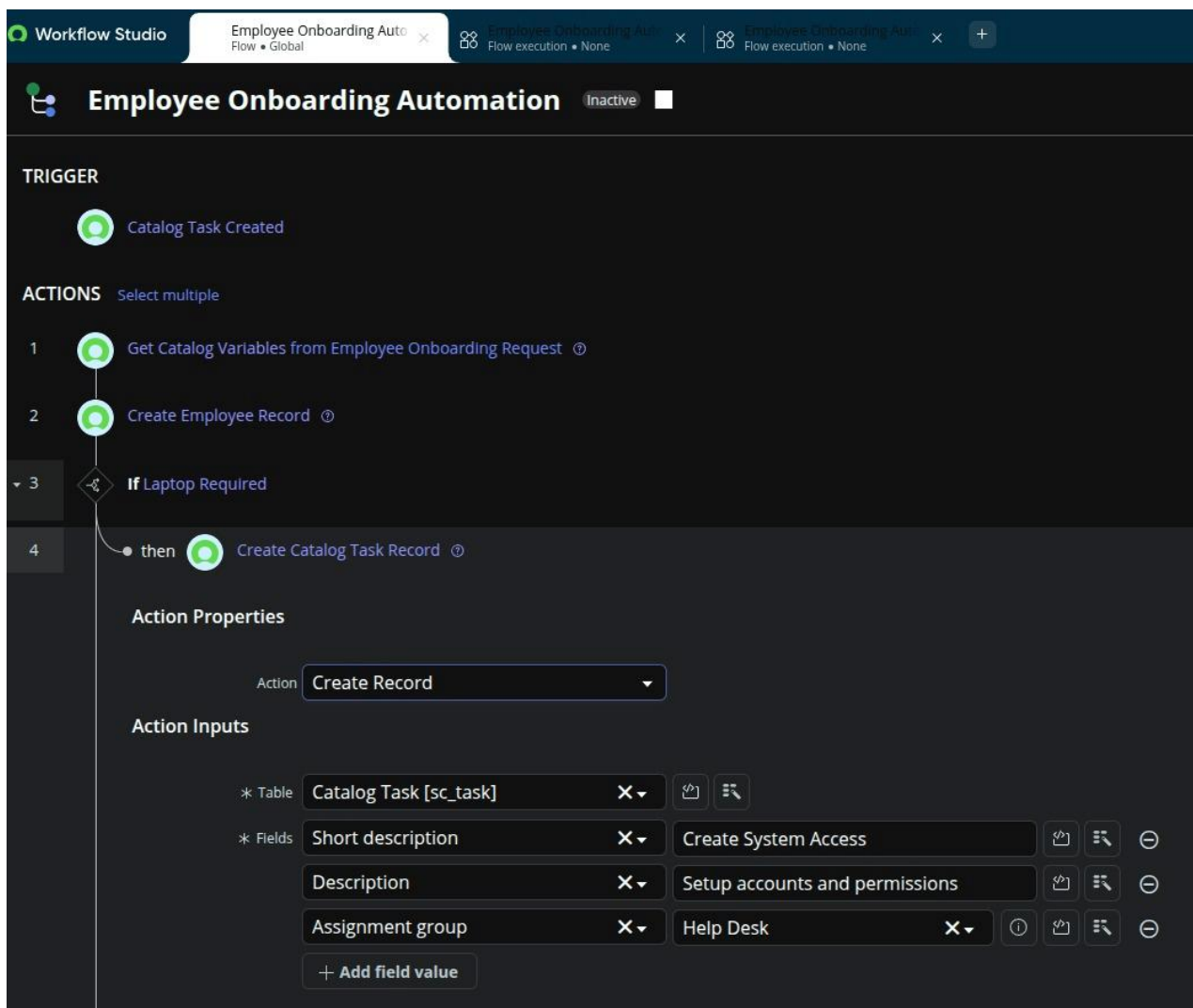
Availability

* Question: Manager

* Name: manager

Conversational label:

Tooltip:



Conclusion:

The Automated Employee Onboarding & Offboarding System in ServiceNow successfully transforms traditional manual employee lifecycle processes into a streamlined, automated, and secure digital workflow. The solution enables organizations to efficiently manage employee onboarding and offboarding activities by automating request submission, manager approvals, departmental task assignments, SLA monitoring, and status tracking.

By leveraging Service Catalog, Flow Designer, custom Employee Lifecycle tables, role-based ACLs, notifications, and reporting dashboards, the system improves collaboration between HR, Managers, IT, Facilities, and Security teams while reducing manual effort and process delays. Dynamic forms and automated workflows ensure that users provide accurate information, while role-based access controls protect sensitive employee data and maintain compliance.

The implementation provides complete lifecycle visibility through centralized data management, audit tracking, and performance reports. QA validation confirmed that request creation, approval routing, task assignment, notifications, SLA tracking, and record closure work as expected.

The project demonstrates how ServiceNow can be used as an enterprise workflow automation platform to enhance employee experience, improve operational efficiency, and support secure lifecycle management. Future enhancements can include integrations with external HRMS platforms, Identity and Access Management systems, and Asset Management solutions to achieve fully automated end-to-end employee lifecycle processing.

